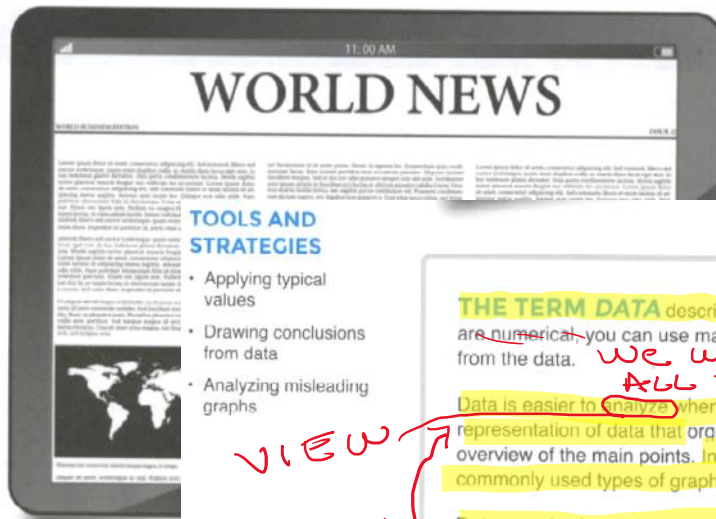




DATA ANALYSIS

SKILLS

- Reading tables and graphs
- Reading histograms and scatter plots
- Finding mean, median, and mode
- Calculating weighted averages
- Reading box plots
- Analyzing distribution
- Seeing trends and making predictions
- Analyzing correlation
- Identifying a representative sample
- Using more than one data source



TOOLS AND STRATEGIES

- Applying typical values
- Drawing conclusions from data
- Analyzing misleading graphs

VIEW

Do NOT say "NORMAL"
Because that is a
special math word.

SET is the math word

THE TERM DATA describes a collection of facts. When the facts are numerical, you can use mathematics to analyze and draw conclusions from the data. we use math to analyze ALL TYPES of data

Data is easier to analyze when it is organized. A **graph** is a visual representation of data that organizes the information and provides a quick overview of the main points. In this unit, you'll study several of the most commonly used types of graphs.

Data can also be easier to analyze when it is summarized. **Typical values** represent what is normal or average, in a set of data. You'll learn three types of typical values and how to use them to draw conclusions. mean median mode

You can also use patterns in the data to draw conclusions. In this unit, you'll learn to recognize patterns and make predictions based on those patterns. SAME THING AS

Data Analysis

Wednesday, December 11, 2024 11:45 AM

USING DATA ANALYSIS

You use data analysis when you:

- Read a table to find out how much medicine to give a child ✓
- Analyze sports statistics to pick the winner of a game or event ✓
- Interpret a graph in a magazine or newspaper ✓
- Decide whether to believe the information in a sales pitch ✓
- Read a bus or train schedule ✓

Read the following items.

1. Look through a newspaper or a magazine and count how many graphs you find. Which graph is easiest to read? Why?

on

2. Have you ever calculated your own grade in a class or your grade point average (GPA)? How do you determine what your average should be?

This is called a "weighted average." For example, suppose you are taking two courses, one of which is 3 credit hours and the other is 4 credit hours. Suppose further that points are assigned as follows: A - 4 pts, B - 3, C - 2, D - 1, F - 0. Further suppose you earned an A for the 3 credit hour course and a C for the 4 hour course. Then your "weighted average" is:

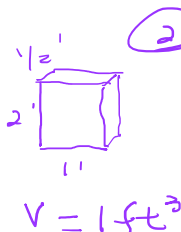
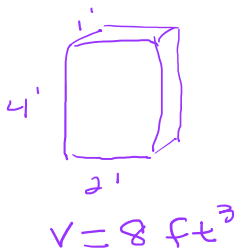
$$[(3 \text{ credit hours})(4 \text{ points per credit hour}) + (4 \text{ credit hours})(2 \text{ points per credit hour})] / 7 \text{ hours total} = 2.85 \text{ points/hour} . \text{ GPA} = 2.85$$

3. Think of some things that increase when the temperature increases. For example, when the temperature increases, the number of people at a water park increases. What things decrease when it gets hot out?

Heating Bill decreases
In Northern Hemisphere, less lights are used .

4. Do you trust the graphs you see in newspapers or magazines or on the Internet? What makes a graph trustworthy? What makes it misleading?

① GRAPHS must start at zero .



② Improper Graphs .
The left looks Double the Right ,
except its volume is
8 times greater !!

MATH TALK

In real life, people are not usually interested in an entire set of data. They want to know what's ~~real~~, usual, or typical. For example, if someone asks you about how much a large pizza costs, they probably don't want to know the price of a large pizza at every restaurant in the city. They just want to know that a large pizza normally costs about \$~~13.00~~ ^{\$13.00}. They want to know the typical value.

With a partner, make a list of situations where typical values are all you really need to know. Then make a list of situations where a typical value isn't enough information.

TYPICAL
IS ENOUGH

- * Rent of a 1-Bedroom apartment
- * Cost of a week's groceries for a family of four
- * Cost of 2 days' hospital stay for an appendectomy

TYPICAL
IS NOT ENOUGH

- * Cost of Carpet Cleaning Based on Square Footage
- * Cost for a set of 100% cotton sheets (quality differs greatly among sellers)

WORKING WITH TABLES

A **table** is often used to organize and display data, especially data that is classified by categories. A table may contain both words and numbers written in labeled rows and columns.

A **row** is read from left to right. A **column** is read from top to bottom.

Use the row and column headings to locate specific data in a table. The title of the table may also have important information.

COLUMNS

Population of the Top 5 U.S. Cities (2000 and 2010)
(in millions)

ROWS
→

City	2000 Population	2010 Population	2010 Rank	Percent Change 2000-2010
Chicago, IL	2.89	2.69	3	-6.9%
Houston, TX	1.95	2.10	4	+7.5%
Los Angeles, CA	3.69	3.79	2	+2.6%
New York, NY	8.01	8.17	1	+2.1%
Philadelphia, PA	1.51	1.52	5	+0.6%

Source: U.S. Census, 2000 and 2010

qualitative (non-numeric).
data is always organized by categories.

Take a minute to read the table.

Let's check % change:

(start - end)

start

$$\frac{(2.89 - 2.69)}{2.89}$$

$$= \frac{\text{change}}{\text{start}}$$

$$= \frac{0.2}{2.89}$$

$$= \% \text{ change}$$

$$= 6.9\%$$

decrease

$$\text{or } -6.9\%$$

FINDING INFORMATION IN TABLES

Example In 2000, the population of New York City was about how many times larger than the population of Houston? NYC 8.01, Houston 1.95

Locate the place where the column labeled "2000 Population" and the row labeled "New York, NY" intersect.

The value is 8.01. The title states that the values are in millions. The population of New York City was about 8.01 million in 2000.

Now look for the row labeled "Houston, TX." The population of Houston was about 1.95 million in 2000.

$8.01 \div 1.95 \approx 8 \div 2 \approx 4$. The population of New York City was about 4 times larger than the population of Houston in 2000.

	2000 Population
	2.89
	1.95
	3.69
New York, NY	8.01
	1.51

about 4 times larger

$$\frac{8.01 - 1.95}{8.01}$$

$$= 6.06 / 8.01$$

$$= 75.79\%$$

which is correct?
Both!!

To go from 8 to 2, is a decrease of about 75%.

that means 2 is 25% of 8.

∴ 8 is 4 times larger than 2

You can also use the data in a table to draw conclusions. For example, look at the column labeled "Percent Change." Chicago was the only city with a decreasing population. You could conclude that from 2000 to 2010, the populations of the largest U.S. cities primarily increased.

Tables: Examples #1 & #2

Wednesday, December 11, 2024 11:53 AM

Use the following table for problems 1 and 2. *Always take time to analyze the table.*

Summary of Saturday Price Reductions

Item	Original Price	Discount Rate
Portable Speaker	\$80	15%
Earbuds	\$18	5%
MP3 Player	\$26	10%
DVD Player	\$120	25%
HD Television	\$230	20%

Disc Amt	New Price
\$12	\$68
\$0.90	\$17.10
\$2.60	\$23.40
\$30	\$90
\$46	\$184

1. What is the difference in price between a portable speaker and a DVD player on Saturday?

\$68 ✓

\$90 ✓

\$22 ✓

2. Is the actual discount (in dollars) on an HD television greater than or less than the actual discount on a DVD player?

\$46 ✓

\$30 ✓

Greater ✓

① speaker
15% of \$80 is \$12

② earbuds
5% of \$18 is \$0.90

③ MP3
10% of \$26 is \$2.60

④ DVD
25% of \$120 is \$30

⑤ 20% of \$230 is \$46

Tables: Examples #3-6

Wednesday, December 11, 2024 11:54 AM

Use the table for problems 3–6.

Yard Waste Recycled by County (in tons)

Month	Carroll County	Logan County
January	520	1,345
February	1,080	2,100
March	990	2,980
April	2,250	3,405

3. How much more yard waste did Logan County recycle in February than Carroll County?

FEB. $\frac{\text{Logan}}{2100} - \frac{\text{CARROLL}}{1080}$
 1020 tons

5. During which month did Logan County's recycling increase the most compared to the previous month?

FEBRUARY

Jan 1345
Feb 2100 $\rightarrow +755$
Mar 2980 $\rightarrow +880$
Apr 3405 $\rightarrow +425$

was this the largest increase compared to the previous month?

$\frac{\text{Jan} \rightarrow \text{Feb}}{(\text{start} - \text{end}) / \text{start}} = \frac{1345 - 2100}{1345} = \frac{755}{1345} = 56.19\%$

$\frac{\text{Feb} \rightarrow \text{Mar}}{2100 - 2980 / 2100} = \frac{880}{2100} = 41.9\%$

$\frac{\text{Mar} \rightarrow \text{Apr}}{2980 - 3405 / 2980} = \frac{425}{2980} = 14.3\%$

don't worry about the neg's

4. During which month did Carroll County's recycling decrease compared to the previous month?

MARCH

6. From January to April, which county's recycling increased the most, and by how much?

	Carroll	Logan
JAN.	520	1345
APR.	2250	3405
increase	1730 ✓	2060 ✓
% increase	$1730 / 520 = 332.7\%$	$2060 / 1345 = 153.2\%$
check	$520 (332.7\%) = 1730$	$1345 (153.2\%) = 2060$

$\therefore$ by Volume, Logan Cnty

$\therefore$ by Relative Volume, Carroll Cnty

The problem here is Change versus Relative Change. It is absurd to say Logan's recycling increased the most. *Since they don't have the same starting point*, one absolutely must use Relative Change. Therefore, Carroll Cnty had the greater relative change; thus their recycling increased the most, not Logan Cnty.

Tables: Examples #7 & #8

Wednesday, December 11, 2024 11:56 AM

Use the mileage chart for problems 7 and 8.

Airline Mileage to North American Cities

Cities	Mexico City	Montreal	New York	San Francisco
Mexico City		2312	2085	1883
Montreal	2312		330	2537
New York	2085	330		2566
San Francisco	1883	2537	2566	

7. How many miles will Armando travel if he flies from Mexico City to San Francisco and continues to New York?

$$1883 + 2566 = 4449 \text{ miles} \quad \checkmark$$

8. **Explain** Why are there four empty boxes on this chart?

same city to
same city ✓

Quantitative & Qualitative Data

Wednesday, December 11, 2024

11:50 AM

Quantitative Data is numeric data for which math operations can be performed.

For Example:

- Age
- Height
- Weight
- Length
- Volume
- Pressure

Qualitative Data is nonnumeric data for which math operations cannot be formed.

For Example:

- Identity Number
- Street Address
- Movie Titles
- Authors
- Painters
- Colors
- Foods

Notice: *Some quantitative data consists of numbers, but because we can't perform math operations on these numbers (such as your street address), those numbers are considered non-numeric.*

Bar Graphs (Qualitative Data)

Wednesday, December 11, 2024 11:57 AM

BAR GRAPHS AND HISTOGRAMS

Graphs give a **visual picture** of data. In a **bar graph**, data is represented by the height (or length) of **data bars**. The bars may be drawn vertically or horizontally.

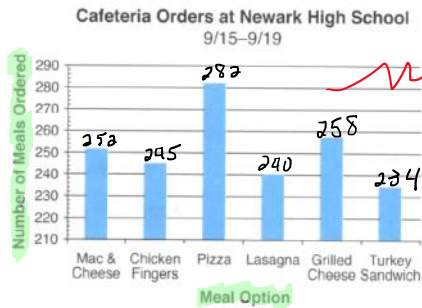
For the bar graph below, you can use the **title** (Cafeteria Orders at Newark High School) and axis labels to determine what values are being compared. The **horizontal axis** shows the type of meal ordered. The **vertical axis** shows the number of meals ordered during the week.

FINDING INFORMATION ON BAR GRAPHS

Example About how many orders of chicken finger meals were sold at Newark High School from 9/15 to 9/19?

Find the bar labeled Chicken Fingers. Read the value on the vertical axis closest to the height of the bar.

You may need to estimate values on a bar graph. The bar for chicken fingers is about halfway between 240 and 250. **About 245 chicken finger meals** were sold from 9/15 to 9/19.



Qualitative Data only
(non-numeric)

notices spaces
Between Bars

which is the Data?

- meal Option?

- # meals?

→ these are
simply the counts

Use the bar graph above for problems 1 and 2.

1. Which meal options sold over 250 meals from 9/15 to 9/19?

Mac & Cheese ✓
Pizza ✓
Grilled Cheese ✓

2. About how many more orders were for pizza than for lasagna?

282 ✓
240 ✓
42 ✓

answer key → 280

Bar Graphs: Examples

Wednesday, December 11, 2024 11:58 AM

Use the double bar graph for problems 3–6.

3. About how much more did the petting zoo make in Year 2 than Year 1?

$$\$4000 - \$2500 = \$1500 \checkmark$$

4. In which event did profits decrease from Year 1 to Year 2?

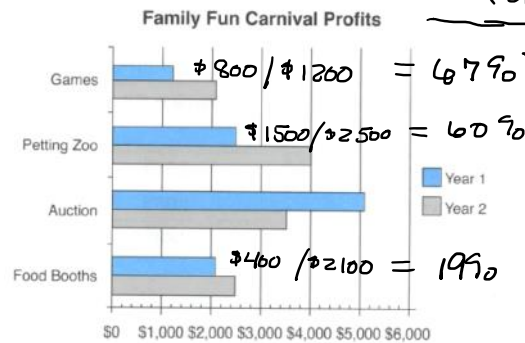
Auction $\checkmark$

5. In Year 1, about how much more money did the auction make than the food booths?

$$\$5100 - \$2100 = \$3000 \checkmark$$

6. Which event's profits increased the most from Year 1 to Year 2?

Exact Profits: Petting Zoo @ \$1500 $\checkmark$
Relative Increase: Games @ 67%



for question #6

If I were the owner of the Carnival, I would award the Manager of the Games the biggest % raise in salary.

Histograms (Quantitative Data)

Wednesday, December 16, 2020 12:01 PM

Remember: Bar graph - qualitative data
Histogram - quantitative data

numeric data only

A histogram also uses data bars to visually show data. On a bar graph, the bars represent categories. On a histogram, the bars represent a range of numbers. You can use a histogram to draw conclusions about how frequently events occur.

Each bar on the histogram below represents an interval. For example, the first bar represents the number of cars sold with an estimated gas mileage of 5 to 9 miles per gallon. Use the vertical axis to read the value of the first bar. Two cars were sold with an estimated gas mileage between 5 and 9 miles per gallon.

FINDING INFORMATION ON HISTOGRAMS

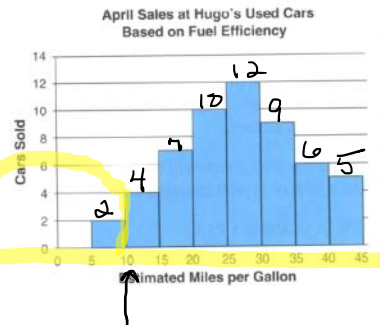
Example How many cars were sold in April with an estimated gas mileage of less than 15 miles per gallon?

Find 15 on the horizontal axis. The bars to the left of this marker represent cars with a mileage of less than 15 mpg.

Read the values for both bars and find the total.

5 to 9: 2
10 to 14: 4
Total: $2 + 4 = 6$

In April, 6 cars with an estimated gas mileage of less than 15 mpg were sold.



Use the histogram above for problems 7 and 8.

7. Did Hugo's Used Cars sell more cars with an estimated mileage greater than 25 mpg or less than 25 mpg?

8. How many cars did Hugo's Used Cars sell in April?

$$23 + 32 = 55 \checkmark$$

$$< 25: 2 + 4 + 7 + 10 = 23 \checkmark$$

$$> 25: 12 + 9 + 6 + 5 = 32 \checkmark$$

Note - the way the x-axis of the graph is written, it is impossible to answer the question: How many cars had an estimated mileage of 25 mpg?

could be a range such as GROUPED

10-4 | 5-9 | 10-14 ...

or single values UNGROUPED

10 | 1 | 2 | 3 | ...

STAT / WORDS

HORRIBLY BAD GRAPH !!

* what represents 10?

0 25 10 15 20 etc.

OR 0-9 5-9 10-14 etc

OR 5 10 15 20 ...

Histograms: Examples

Wednesday, December 11, 2024 12:02 PM

Use this histogram for problems 9 and 10.

9. A grade below 70% is a failing grade. How many students failed the final?

$$1 + 1 + 2 + 4 = 8 \quad \checkmark$$

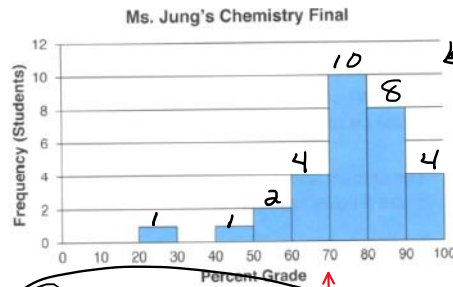
10. Explain Students who scored above 70% passed the final. To the nearest percent, what percent of Ms. Jung's class passed the final? Explain your answer.

$$\begin{aligned} \# \text{ passed} &= 10 + 8 + 4 \\ &= 22 \end{aligned}$$

$$\begin{aligned} \text{tcl \#} &= 8 + 22 \\ &= 30 \end{aligned}$$

$$\% \text{ passed} = \frac{22}{30}$$

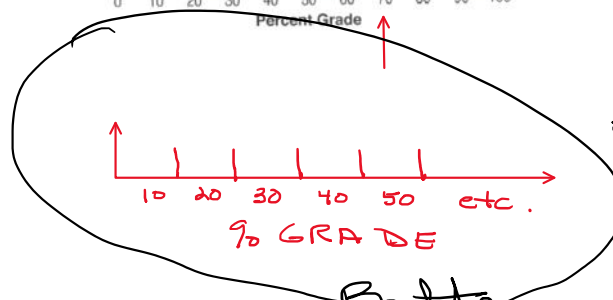
$$= 73.\bar{3} \% \quad \checkmark$$



cannot tell exactly # students who earned a 50%, or a 60%, etc.

scratch

$$\begin{array}{r} 73 \\ 30 \overline{) 2200} \\ \underline{210} \\ 100 \\ \underline{90} \end{array}$$



Better x-axis